Proceedings of the 2026 Winter Simulation Conference
V. Ramamohan, A. Djanatliev, M. Fakhimi, C. Krejci, C. Ruiz Martin, B. S. Onggo, and N. Mustafee, eds.

HEAS: A HIERARCHICAL EVOLUTIONARY AGENT-BASED SIMULATION FRAMEWORK FOR MULTI-OBJECTIVE POLICY SEARCH

Ruiyu ZHANG[1], Lin NIE[2], and Xin ZHAO[2]

[1]Dept. of Politics and Public Administration, The University of Hong Kong, Hong Kong SAR, CHINA
[2]Dept. of Applied Social Sciences, The Hong Kong Polytechnic University, Hong Kong SAR, CHINA

ABSTRACT

HEAS is a reusable framework for simulation-based policy search that standardizes metric extraction across evolutionary search, tournament evaluation, and statistical validation. The framework targets a recurring ABM integration problem: researchers often rebuild 100–200 lines of coupling code per project, and the same outcome metric can silently diverge across optimization, comparison, and inference code paths. HEAS addresses this through four commitments: a Layer/Stream/Arena composition model; a uniform metric contract consumed identically by the evolutionary algorithm, tournament scorer, and bootstrap CI pipeline; native NSGA-II integration; and a Tournament primitive with four voting rules and noise stability analysis. Against a Mesa 3.3.1 baseline, HEAS requires 5 coupling lines versus 160 (97% reduction). Three case studies—ecological, enterprise, and a Wolf-Sheep mean-field reimplementation—demonstrate the framework across two simulation scales. In the tested ecological scenarios, the evolved champion wins all 32 held-out cases and all 24 single-anchor OOD cases.

1 INTRODUCTION

HEAS addresses a recurring integration problem in simulation-based policy search: the same metric is often recomputed in incompatible ways across optimization, comparison, and inference. In a representative Mesa workflow, this appears as roughly 160 lines of coupling code; in HEAS, the same workflow requires five lines because all downstream stages consume one metric contract.

Agent-based models (ABMs) have become a primary tool for studying emergent behavior in complex adaptive systems—from ecological communities and financial markets to public health interventions and regulatory policy design. Using an ODE-style or tabular ABM for *policy search* requires integration infrastructure that no general-purpose ABM framework provides as a reusable layer. HEAS targets this specific niche—coupled ODE/tabular dynamics, multi-objective EA, multi-run statistics—rather than the full ABM space including spatially-explicit heterogeneous-agent systems.

1.1 The Coupling Code Problem

Consider a researcher using Mesa to study regulatory policy in a multi-firm economy. She wants to (1) evolve a Pareto-optimal tax regime using NSGA-II, (2) compare the evolved regime against a reference policy across independent economic scenarios using tournament voting, and (3) report bootstrap confidence intervals. Each step reads the same welfare metric, yet in Mesa each is an independent code path: the DEAP fitness function extracts welfare from the model's `DataCollector` after each run; the tournament scorer extracts welfare from a separate episode-evaluation function, possibly with different aggregation; and the bootstrap CI loop extracts welfare from a third data collection loop, potentially inconsistent with both.

When the researcher adds a second objective—welfare inequality—she must edit the DataCollector reporter, the DEAP fitness function, the tournament scorer, and the CI loop, keeping all four in sync manually. A silent divergence can persist undetected for weeks and invalidate a publication. This pattern,

which we call *coupling code*, appears in representative ABM-based policy-search workflows (Pangallo, Farmer, Sanders, and Galla 2019; Caprì, Ignaccolo, Inturri, Pluchino, and Rapisarda 2023).

1.2 Contributions

The paper's primary contribution is a reusable framework architecture centered on a uniform metric contract; the optimizer and tournament modules are reusable implementations of that contract, and the case studies serve as validation of reuse and auditability. The design has two central elements: a `Layer`/`Stream`/`Arena` composition model for structuring simulations, and a uniform `metrics_episode()` contract that is consumed identically by optimization, tournament scoring, and statistical inference. The optimizer, tournament, and statistics modules are reusable implementations built on that contract.

Section 5 evaluates this framework claim through coupling-code reduction, two-scale ecological studies, tournament stability, and cross-domain portability.

2 RELATED WORK

Agent-based modeling is widely used to study complex adaptive systems in ecology (Grimm, Berger, Bastiansen, Eliassen, Ginot, Giske, Railsback, and DeAngelis 2006), computational economics (Tesfatsion 2006), and organizational behavior (Bonabeau 2002). Macal and North (Macal and North 2010) identify a recurring architectural division: frameworks such as Mesa (Kazil, Masad, and Crooks 2020) and NetLogo (Wilensky 1999) support model execution but provide no native policy-search infrastructure. The ODD protocol (Grimm, Berger, Bastiansen, Eliassen, Ginot, Giske, Railsback, and DeAngelis 2006) advances ABM reproducibility through model description, but does not address the statistical machinery needed for multi-run policy comparison.

Simulation optimization has a rich tradition at WSC (Fu 2002; Hong and Nelson 2009). Fu (Fu 2002) identifies the persistent theory-practice gap between optimization formulations and the coupling code required to connect solvers to simulation models. Hong and Nelson (Hong and Nelson 2009) provide a practitioner-oriented introduction that highlights the per-project engineering burden. The SimOpt testbed (Pasupathy and Henderson 2011) catalogs benchmark problems, while commercial platforms such as OptQuest (OptTek Systems, Inc. 2007) and Pagmo (Biscani and Izzo 2020) supply search backends—yet none address the metric-contract problem specific to multi-objective ABM workflows. Multi-objective simulation optimization has received growing WSC attention (Hunter and Pasupathy 2022), and evolutionary approaches including NSGA-II (Deb, Pratap, Agarwal, and Meyarivan 2002) are increasingly deployed in agent-based settings (Caprì, Ignaccolo, Inturri, Pluchino, and Rapisarda 2023).

Prior ABM policy-search studies (Tesfatsion 2006; Pangallo, Farmer, Sanders, and Galla 2019; Zheng, Trott, Srinivasa, Parkes, and Socher 2022) bridge the framework-optimizer gap with bespoke coupling code. Ranking and selection methods (Goldsman and Nelson 1994) address experiment design but assume single-objective outcomes. Optimization libraries such as DEAP (Fortin, De Rainville, Gardner, Parizeau, and Gagné 2012), Optuna (Akiba, Sano, Yanase, Ohta, and Koyama 2019), and Pagmo (Biscani and Izzo 2020) provide flexible search backends but define no metric interface shared across optimization, tournament evaluation, and statistical inference. HEAS's `metrics_episode()` contract closes this gap by being consumed identically at all three stages.

HEAS is adjacent to, but distinct from, benchmark suites, optimizer wrappers, model adapters, and reproducibility tooling: it standardizes the metric pathway shared by optimization, tournament scoring, and inference.

3 THE HEAS FRAMEWORK

HEAS is implemented in Python 3.11 and depends on DEAP for evolutionary algorithms, NumPy for numerical computation, and SciPy for statistical utilities. All components—EA algorithm, voting rule, statistical estimator—can be replaced without changing simulation or metric contract code.

3.1 Layered Composition

This subsection defines the compositional boundary that lets models be reused without rewriting evaluation code. The fundamental abstraction is a `Layer` with four methods (`step`, `metrics_step`, `metrics_episode`, `reset`). All downstream components read the same `metrics_episode()` keys.

Layers compose into **Streams**—named groups forming a coherent sub-simulation—and Streams compose into an **Arena**, which orchestrates step ordering, inter-stream signal routing, and episode-level aggregation. The naming convention `stream_name.metric_name` ensures globally unambiguous keys regardless of Arena size. Explicit composition boundaries make components testable and replaceable without rewriting optimization or tournament code; the uniform metric contract (C2) remains stable as model internals evolve.

3.2 Native Multi-Objective EA Integration

This subsection shows how the metric contract is consumed by optimization without introducing a separate evaluation interface. The wrapper handles DEAP's creator/toolbox registration, crossover and mutation wiring, hall-of-fame management, and hypervolume computation. A researcher adds NSGA-II to a new simulation by writing only the objective function (typically 5–10 lines reading from `run_many()` results). The EA algorithm is replaceable: `run_ea()` also accepts CMA-ES and hillclimbing. Section 5.5 shows that at 1,000-step ecological scale, NSGA-II's advantage over random search is large ($d = 1.39$), showing that algorithm-agnosticism and algorithm quality are complementary properties.

`run_many()` executes n episodes in parallel via `ProcessPoolExecutor`, with per-episode seeds derived deterministically from a base seed, making individual fitness calls reproducible.

3.3 Tournament Evaluation

This subsection defines the comparison layer that uses the same metric contract for multi-scenario policy ranking. Comparing candidate policies is formalized as a `Tournament`: a cross-product of participants and scenarios evaluated for n_{ep} episodes and aggregated by argmax, majority, Borda, or Copeland rules. The rules are intended to reveal whether rankings are robust to aggregation choice. Universal agreement (as in Section 5.4) indicates clear dominance; disagreement diagnoses the competitive structure. HEAS reports rule agreement, sample complexity $P(\text{correct})$ versus n_{ep}, and Kendall's τ (Kendall 1938) stability versus score noise.

3.4 Statistical Infrastructure

This subsection makes the inference layer explicit: the same episode metrics used by optimization and tournaments also feed uncertainty quantification. `heas/utils/stats.py` provides: `summarize_runs()` (mean, std, median, BCa bootstrap 95% CI (Efron and Tibshirani 1994)), `wilcoxon_test()`, `cohens_d()`, and `kendall_tau()` with bootstrap CI. `heas/utils/pareto.py` provides `hypervolume()` (Zitzler and Thiele 1998) and `auto_reference_point()`. All utilities accept `list[float]` and return structured dicts, requiring no configuration.

4 CASE STUDIES

Three case studies span ecological, economic, and published-model domains.

4.1 Ecological Population Management

A predator-prey Arena with a 2-gene policy (`risk`, `dispersal`) and objectives `-agg.mean_biomass`, `agg.cv`. Five streams model climate forcing, landscape quality, prey logistic growth, predator dynamics, and cross-stream aggregation; eight fragmentation $\times$ shock scenarios form the evaluation grid. Section 5.3 validates this model at two simulation scales; the evolved champion is subjected to a 32-scenario out-of-distribution robustness test in Section 5.4.

4.2 Enterprise Regulatory Design

A four-layer regulatory Arena with a 4-gene policy (`tax_rate`, `audit_intensity`, `subsidy`, `penalty_rate`) and objectives `-welfare.total`, `welfare.gini`. Firms maximize per-step profit against a stochastic demand schedule; the regulator, government, and welfare layers implement enforcement, redistribution, and aggregation. The Pareto trade-off is aggregate welfare versus distributional equality across 32 governance scenarios. Section 5.6 reports cross-domain validation at large scale.

4.3 Wolf-Sheep Predation: Mean-Field Reimplementation

To demonstrate compatibility with *existing published models*, we re-implement the canonical Mesa Wolf-Sheep Predation model (Wilensky 1997; Kazil, Masad, and Crooks 2020)—the flagship example distributed with the Mesa library—as a mean-field ODE Arena. Its published dynamics are Lotka-Volterra (Lotka 1925; Volterra 1926) predator-prey with grass regrowth on a 20×20 spatial grid.

HEAS reimplementation. We implement the mean-field ODE approximation as a 4-layer Arena, deriving each term directly from Mesa's energy mechanics. All six published parameters are used verbatim.

Grass. Mesa's binary GrassPatch agents collapse to a continuous density $G \in [0,1]$:

$$\Delta G = \frac{1-G}{\tau_g} - \frac{SG\gamma}{N}, \tag{1}$$

where $\tau_g = 30$, $N = 400$, and $\gamma \in [0,1]$ is the policy grazing-rate gene.

Sheep ($g_s = 4$, $r_s = 0.04$):

$$\Delta S = \Big(r_s - \frac{\max(0,\, 1 - g_s G\gamma)}{g_s}\Big) S \;-\; \frac{WS}{N}. \tag{2}$$

Wolves ($g_w = 20$, $r_w = 0.05$; h = harvest-rate gene):

$$\Delta W = r_w \frac{WS}{N} \;-\; \frac{\max(0,\, 1 - g_w S/N)}{g_w} W \;-\; hW. \tag{3}$$

Every coefficient maps directly to a published Mesa default; no parameters are ad-hoc. Concretely: `grass_regrowth_time` $\to \tau_g$, `sheep_gain_from_food` $\to g_s$, `sheep_reproduce` $\to r_s$, `wolf_gain_from_food` $\to g_w$, `wolf_reproduce` $\to r_w$, `initial_population` $\to N$ (density normalizer). The spatial grid is collapsed to a density field, an acknowledged simplification (Section 6). Adding EA optimization to this published model required zero additional coupling code: the 8-line objective function reads `metrics_episode()` keys, identical in structure to the other two case studies.

A 30-run algorithm comparison on this model is reported in Section 5.5, where the landscape structure determines which optimizer wins.

5 EVALUATION

5.1 Evaluation Questions and Protocol

We pre-specify five evaluation questions covering framework overhead reduction, reproducibility across scales, tournament stability and OOD behavior, algorithm matching, and code-level reuse across heterogeneous tested domains. RQ1 tests overhead reduction; RQ2–RQ4 use the ecological and Wolf-Sheep studies; RQ5 uses separately labeled ecological and enterprise portability runs. Each reported result is tied to one artifact family in the appendix source list.

5.2 Mesa Comparison: Coupling Code Reduction

We implement the ecological case study as a competent Mesa 3.3.1 model and count non-blank, non-comment coupling code (model dynamics excluded). Measured totals: 160 LOC (Mesa) vs. 5 LOC (HEAS)—155 lines saved (97%).

Table 1: Coupling code: Mesa 3.3.1 vs. HEAS vs. Mesa+utility-library (^{est} = estimated; Mesa from `mesa_eco.py`).

Task	Mesa	Mesa+Util	HEAS
Metric contract / DataCollector setup	15	5^{est}	0
Episode metric extraction	20	5^{est}	0
Sequential episode runner	22	3^{est}	1
Parallel runner (`ProcessPoolExecutor`)	20	5^{est}	0
EA fitness glue	14	5^{est}	3
Tournament scorer	18	5^{est}	0
Per-run seed management (30-run)	12	3^{est}	0
Bootstrap CI	35	5^{est}	0
Adding a second objective (incremental)	4	2^{est}	1
Total	**160**	**~38^{est}**	**5**

Listing 1 reproduces the two most illustrative Mesa coupling blocks verbatim, allowing verification without repository access. Adding a metric requires editing Block A *and* every downstream consumer; in HEAS this is a one-line change in one file.

Listing 1: Mesa coupling code Blocks A and B (35 LOC from `experiments/mesa_eco.py`). HEAS equivalent: zero LOC.

```
# Block A - DataCollector setup (15 LOC)
# Adding a metric: edit here AND the EA function
# AND the tournament scorer, all in sync.
self.datacollector = mesa.DataCollector(
   model_reporters={
      "prey":   lambda m: m.prey,
      "pred":   lambda m: m.pred,
      "extinct": lambda m: float(m.pred <= 0.0),
      "biomass": lambda m: m.prey + m.pred,
   }
)

# Block B - episode metric extraction (20 LOC)
# Every consumer (EA, tournament, CI) calls this
# separately. Silent divergence risk.
def extract_episode_metrics(model):
   df = model.datacollector.get_model_vars_dataframe()
   mean_biomass = float(df["biomass"].mean())
   cv = df["biomass"].std() / mean_biomass \
       if mean_biomass > 0 else 0.0
   return {"mean_biomass": mean_biomass, "cv": cv,
         "extinct": float(df["extinct"].iloc[-1])}
```

Table 1 includes a Mesa+utility-library estimate ($\sim$38 LOC). HEAS's claim over this baseline is that C2 enforces the metric interface at the *framework* level, so new projects inherit it without a utility module.

5.3 Multi-Scale Reproducibility

For RQ2, we test whether the ecological study is reproducible across the small- and large-scale settings used in this paper. At small scale ($n = 30$, pop = 20, ngen = 10, steps = 150), ecological HV = 7.665 (std 3.518, CI [6.424, 8.914]). The distribution is bimodal, and a 4×4 budget ablation ($n = 20$ runs/cell; S2c) identifies

population size as the controlling factor: escape rate (HV >9.0) rises from 15–25% at pop ≤ 20 to 40–80% at pop ≥ 30.

The study was then replicated at larger scale (steps = 1,000, n_eval = 5 episodes per genome evaluation, pop = 30, ngen = 15, $n=20$ runs). At this horizon NSGA-II outperforms random search: mean HV = 16.66 (std 10.84, CI [12.37, 20.94]) versus random mean = 6.00 (std 0.34, CI [5.85, 6.14]). Wilcoxon test: $p=1.91\times10^{-6}$; Cohen's $d=1.39$ (large effect); NSGA-II achieves 178% higher mean HV than random search. At 150-step scale the same comparison is not significant ($d=0.49$, $p=0.49$); the Pareto landscape becomes meaningfully richer only at the longer horizon.

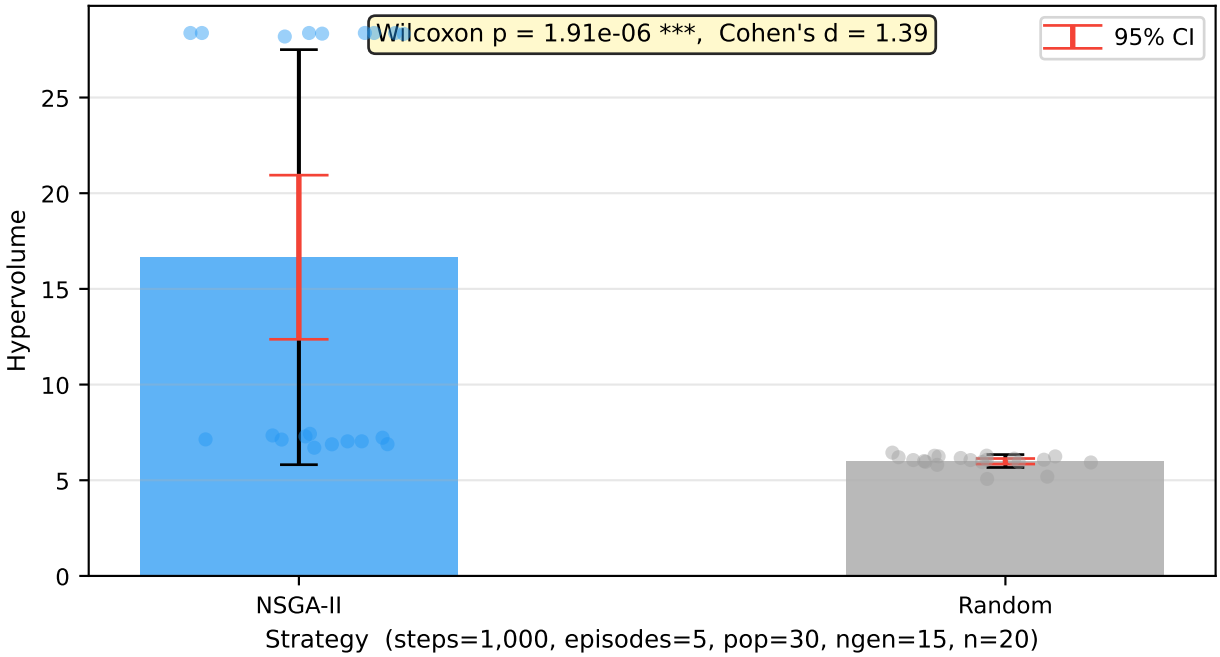

Figure 1: Large-scale algorithm showdown: NSGA-II vs. Random at 1,000 steps/episode ($n=20$ runs). Bars show mean $\pm$ 1 std; error bars show 95% CI; dots show individual runs. Cohen's $d=1.39$; Wilcoxon $p=1.91\times10^{-6}$.

Figure 1 shows per-run distributions and aggregate statistics. The HEAS framework required no code changes between the 150-step and 1,000-step studies: only the `steps` argument changed.

5.4 Tournament Validation and OOD Champion Robustness

For RQ3, we evaluate stable tournament behavior and champion performance in the tested ecological scenario families. Voting-rule agreement and sample complexity were assessed across 30 repeats of the 8-scenario tournament with 100 episodes per scenario. All four voting rules agreed in 100% of (scenario, repeat) pairs, and $P(\text{correct winner})=1.0$ at all tested episode budgets (4, 10, 25, 50, 100 per scenario); a near-tie sensitivity study (S9) confirms that this agreement disappears in close contests.

Noise stability was assessed by injecting Gaussian noise $\mathcal{N}(0,\sigma^2)$ into episode scores (30 repeats) and measuring Kendall's τ (Kendall 1938) between clean and noisy rankings. Rankings are stable ($\tau=1.0$) for noise up to 0.65% of the inter-policy margin: $\tau=0.944$ at $\sigma=10$ (6.5% of margin), degrading to $\tau=0.744$ at $\sigma=50$.

OOD champion robustness was validated at two scales. At small scale (16 scenarios, steps = 500), the champion won all 16 ($p<0.001$). At large scale (32 scenarios), the champion [0.000107, 0.9335] won all

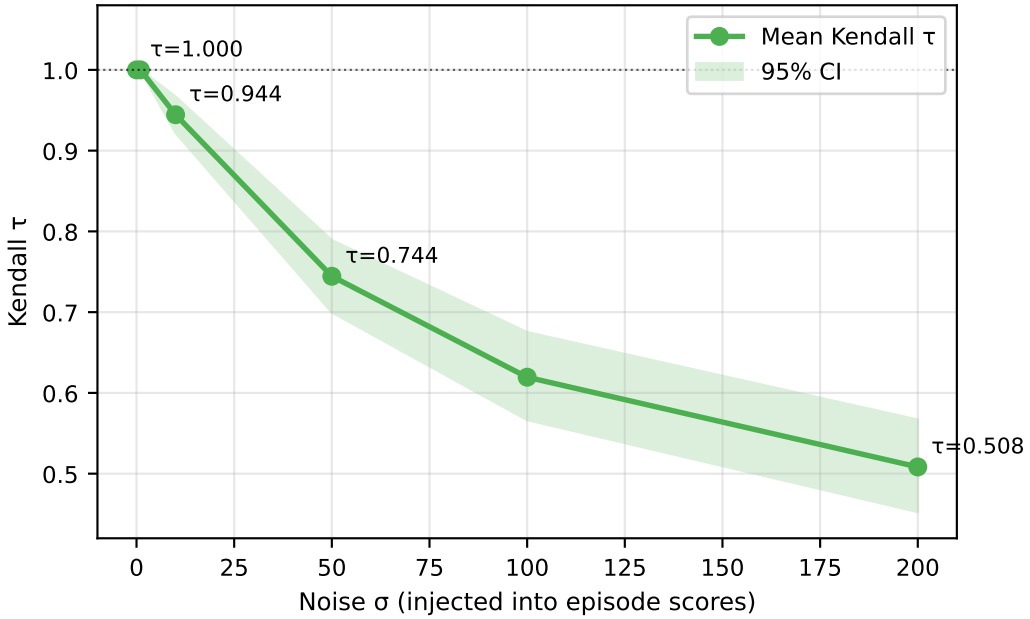

Figure 2: Kendall's τ versus injected score noise σ ($n = 30$ repeats, 8 scenarios). Rankings are stable ($\tau = 1.0$) until noise exceeds $\sim$6% of the inter-policy margin.

32 ($\Delta = +25.0$, $+3.96\%$; $p = 4.66 \times 10^{-10}$; $d = 0.17$). In the single-anchor *true OOD* study (Exp. C, S6c), the champion won all 24 scenarios where all four factors lay outside training ($p = 1.19 \times 10^{-7}$), matching the 100% in-distribution win rate. Here win rate and paired deltas are more informative than Cohen's d because the score distributions are nearly deterministic.

5.5 Algorithm Agnosticism and Scale Sensitivity

For RQ4, we test whether HEAS remains useful when the best optimizer depends on landscape structure. An algorithm ablation at steps = 300 (pop = 20, ngen = 10, $n = 10$ runs) finds that Simple hillclimbing outperforms NSGA-II (means: Simple = 19.66, NSGA-II = 9.99, Random = 7.61). On a 2-gene landscape the Pareto front is effectively one-dimensional; single-objective search concentrates population on the biomass-maximizing direction while NSGA-II distributes across the CV trade-off. This reinforces the agnosticism claim: HEAS allows algorithm selection without any framework changes.

Scale sensitivity is reported in Figure 4, which maps mean HV across a 3×3 grid of steps $\in \{140, 300, 500\}$ and episodes per evaluation $\in \{5, 10, 20\}$ ($n = 10$ runs each). HV rises from 6.0 (140 steps, 5 episodes) to 10.1 (300 steps, 20 episodes) within the small-scale regime, confirming that longer episodes yield richer Pareto landscapes. The bimodal variance pattern persists across all cells (consistent with the population-budget ablation findings in Section 5.3).

The Wolf-Sheep algorithm comparison (steps = 200, pop = 50, ngen = 25, $n = 30$ runs) yields Random = 14.35 (std 0.61), NSGA-II = 10.36 (std 4.07), and Simple = 9.14 (std 3.68). Random achieves the highest HV; a two-sided Wilcoxon rank-sum test confirms the difference is significant ($p = 7.99 \times 10^{-6}$, Random > NSGA-II). The Wolf-Sheep Pareto front consists of a narrow band of sustainable harvest-grazing combinations; within this concentrated geometry, exhaustive random exploration dominates NSGA-II's diversity-preserving search.

Algorithm-landscape matching—made practical by C3's pluggable optimizer interface—is the operational takeaway: on simple or single-equilibrium landscapes any search strategy suffices; on complex multi-objective landscapes NSGA-II's structural advantage is large ($d = 1.39$).

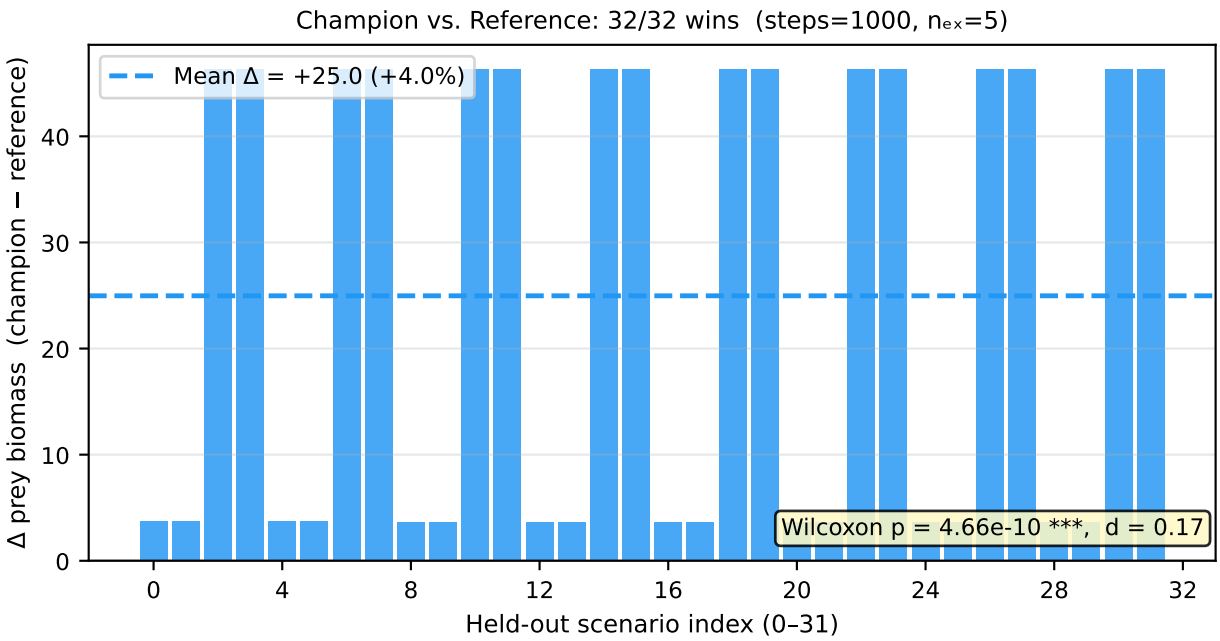

Figure 3: Champion minus Reference biomass across 32 held-out ecological scenarios. Blue bars indicate champion wins (32/32); dashed line marks mean $\Delta = +25.0$ biomass. Wilcoxon $p = 4.66 \times 10^{-10}$.

5.6 Cross-Domain Portability

For RQ5, we ask whether the same framework code can be reused across domains at scale. We report the large-scale ecological and enterprise portability runs from S8, and keep the separate small-scale enterprise validation (S2b) out of this comparison. The same codebase—with no changes to framework code beyond domain-specific `metrics_episode()` implementations—was run on two heterogeneous domains. In the ecological domain (steps = 1,000), NSGA-II achieves mean HV = 15.46 (std 8.57, CI [12.03, 18.88]); the bimodal convergence pattern persists. In the enterprise domain (steps = 200, $n = 20$), NSGA-II achieves mean HV = 5.565×10^6 (std 6.96×10^4, CI $[5.535, 5.594] \times 10^6$). The portability claim is therefore code-level rather than outcome-level: the same framework interfaces support both domains without rewriting optimization, tournament, or statistics code. Tournament validation was also conducted in the enterprise domain across 8 governance scenarios (20 repeats, 30 episodes/scenario) (S10). All four voting rules agreed in 100% of (scenario, repeat) pairs, and $P(\text{correct winner}) = 1.0$ at all tested episode budgets (4, 10, 25, 50), showing that the tournament infrastructure transfers across these tested domains without framework modification.

5.7 Threats to Validity

Construct validity. The case studies are composition vehicles, not calibrated domain models; results are evidence about framework behavior. The metric contract prevents *accidental divergence across consumers*: all three pipeline stages read the same key. An incorrect `metrics_episode()` implementation will be consistently wrong rather than silently inconsistent. The framework additionally exposes `validate_metrics_episode()`, which raises `TypeError` when a `metrics_episode()` implementation returns a non-dict or non-float value, catching contract violations early in development.

Internal validity. Fixed evaluation seeds isolate framework effects but can suppress variance in some metrics; zero-variance outcomes are treated as diagnostic signals (e.g., the $K = 120$ degeneracy, enterprise determinism) rather than performance claims. OOD validation at steps = 500 and on single-anchor

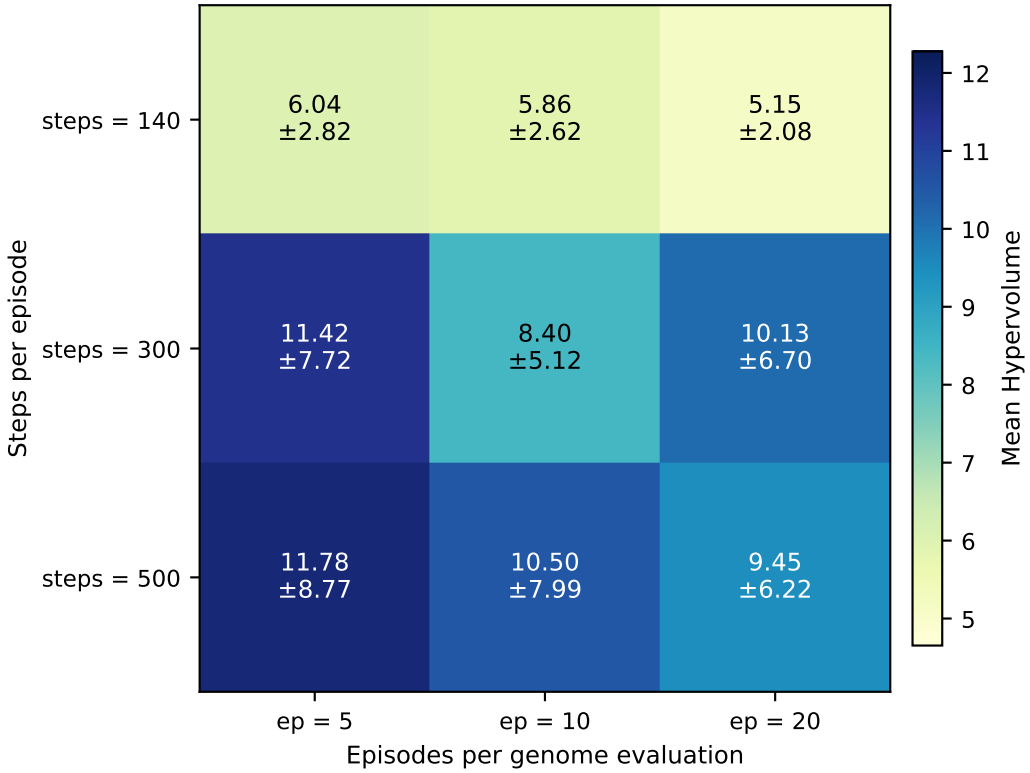

Figure 4: Scale sensitivity heatmap: mean HV by simulation horizon (steps × episodes/genome). Each cell is $n = 10$ independent NSGA-II runs. HV generally increases with steps; non-monotonicities between the 300- and 500-step rows reflect high between-run variance. Episode count has a weaker secondary effect.

extrapolation scenarios (Exp. C) extends evaluation beyond the training horizon, but only within the tested family of ecological scenarios.

External validity. The Mesa baseline is a competent single-project implementation, not a census of all coding styles. Runtime findings for `n_jobs` depend on episode cost and hardware. All three case studies use ODE-style or tabular dynamics; researchers working with genuinely spatially-explicit ABMs—heterogeneous agents on a grid, emergent network effects, rich move-and-interact loops—will encounter integration patterns outside HEAS's current design scope. The 97% LOC reduction is validated specifically for the tabular policy-search workflow that HEAS targets. The results therefore support HEAS on the tested model family, not on arbitrary ABM domains.

6 CONCLUSION

These experiments support a narrower claim: HEAS makes simulation-based policy search more auditable and reusable by enforcing one metric contract across optimization, comparison, and inference. Across five evaluation questions, the paper shows 97% coupling-code reduction against the specific Mesa baseline (160 vs. 5 LOC), reproducible ecological behavior at two scales, stable tournament rankings under realistic noise ($\tau > 0.94$), and algorithm flexibility when landscape structure favors simpler optimizers. Three case studies—including a mean-field compatibility demonstration based on Mesa's published Wolf-Sheep model—show how the same framework interfaces can be reused across the tested model family. Within the tested ecological scenario families, the evolved champion wins all 32 held-out cases and all 24 single-anchor

OOD cases. A 4×4 budget ablation explains the observed bimodal convergence as a diversity-threshold transition around pop = 30.

Limitations. For lightweight ODE simulations (<0.01 s/episode), process startup overhead (≈9 s) dominates; use `n_jobs=1`. HEAS is designed for ODE-style and tabular simulations; spatially-explicit ABMs remain Mesa's domain. The metric contract is scalar-focused (`dict[str, float]`); workflows needing agent-level distributions require an additional aggregation layer.

Future work. Priorities are a `MesaLayerAdapter` (wrapping existing Mesa models as HEAS Layers without rewriting dynamics), scalable parallel backends (Ray/Dask), and automated scenario generation.

For double-blind review, repository access details are omitted. An anonymized artifact package containing all scripts under `experiments/` is prepared for reviewers.

A REPORTED MEANS AND SOURCES

Table 2: Principal results reported in this paper with source file IDs (S1–S10).

Quantity	Value	Source
Coupling LOC reduction	97% (160 vs. 5)	S1
Ecological HV, small scale	7.665 ± 3.518	S2
Enterprise validation HV	4141.74 / +9.5%	S2b
Noise τ at $\sigma=10$	0.944	S3
Noise τ at $\sigma=50$	0.744	S3
Simple ablation HV mean	19.66	S4
NSGA-II ablation HV mean	9.99	S4
Random ablation HV mean	7.61	S4
Ecological HV, large scale	16.66 ± 10.84	S5
Random HV, large scale	6.00 ± 0.34	S5
NSGA-II vs. Random d	1.39	S5
16-scenario OOD wins	16/16 ($p<0.001$)	S6a
32-scenario OOD wins	32/32 / +25.0 (+3.96%)	S6b
True OOD 24-scenario wins	24/24 / 1.19×10^{-7}	S6c
Wolf-Sheep NSGA-II vs. Random p-value	7.99×10^{-6}	S7
Enterprise portability HV	5.565×10^{6}	S8
Near-tie mean off-diagonal agreement	0.755 (24.5% disagree)	S9
Enterprise tournament agreement (100%)	100% / $P=1.0$	S10

Source IDs (paths under `experiments/results/`):

S1 mesa_vs_heas/loc_comparison.json
S2 eco_stats/pop20_ngen10_trait/summary.json
S2b ent_stats_small_n20/summary.json
S2c bimodal_ablation/ablation_summary.json
S3 tournament_stress/noise_stability.json
S4 baseline_comparison/ablation.json
S5 large_scale_comparison/p1_summary.json
S6a baseline_comparison/champion_vs_ref.json
S6b large_scale_comparison/p3_summary.json
S6c true_ood/phase2_summary.json
S7 wolf_sheep_study/algo_comparison.json
S8 large_scale_comparison/p2_summary.json
S9 near_tie_tournament/best_case.json
S10 enterprise_tournament/summary.json

REFERENCES

Akiba, T., S. Sano, T. Yanase, T. Ohta, and M. Koyama. 2019. "Optuna: A Next-generation Hyperparameter Optimization Framework". In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2623–2631: ACM.

Biscani, F., and D. Izzo. 2020. "A parallel global multiobjective framework for optimization: Pagmo". *Journal of Open Source Software* 5(53):2338.

Bonabeau, E. 2002. "Agent-Based Modeling: Methods and Techniques for Simulating Human Systems". *Proceedings of the National Academy of Sciences* 99(Suppl 3):7280–7287.

Byrd, D., M. Hybinette, and T. H. Balch. 2020. "ABIDES: Towards High-Fidelity Multi-Agent Financial Market Simulation". In *Proceedings of the 2020 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS)*, 11–22: ACM.

Caprì, S., M. Ignaccolo, G. Inturri, A. Pluchino, and A. Rapisarda. 2023. "Evaluating Transport Policies Through NetLogo Multi-Agent Simulation and Evolutionary Algorithms". *Transportation Research Procedia* 69:613–620.

Collier, N., and M. J. North. 2022. "Repast4Py: Python-based Large-Scale Agent-Based Modeling". In *Proceedings of the 2022 Winter Simulation Conference*, 192–203: IEEE Press.

Deb, K., A. Pratap, S. Agarwal, and T. Meyarivan. 2002. "A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II". *IEEE Transactions on Evolutionary Computation* 6(2):182–197.

Efron, B., and R. J. Tibshirani. 1994. *An Introduction to the Bootstrap*. New York: Chapman and Hall/CRC.

Fortin, F.-A., F.-M. De Rainville, M.-A. Gardner, M. Parizeau, and C. Gagné. 2012. "DEAP: Evolutionary Algorithms Made Easy". *Journal of Machine Learning Research* 13:2171–2175.

Fu, M. C. 2002. "Optimization for Simulation: Theory vs. Practice". *INFORMS Journal on Computing* 14(3):192–215.

Goldsman, D., and B. L. Nelson. 1994. "Ranking, Selection and Multiple Comparisons in Computer Simulation". In *Proceedings of the 1994 Winter Simulation Conference*, 192–199: IEEE Press.

Grimm, V., U. Berger, F. Bastiansen, S. Eliassen, V. Ginot, J. Giske, *et al*. 2006. "A Standard Protocol for Describing Individual-Based and Agent-Based Models". *Ecological Modelling* 198(1–2):115–126.

Hong, L. J., and B. L. Nelson. 2009. "A Brief Introduction to Optimization via Simulation". In *Proceedings of the 2009 Winter Simulation Conference*, 75–85: IEEE Press.

Hunter, S. R., and R. Pasupathy. 2022. "Central Limit Theorems for Constructing Confidence Regions in Strictly Convex Multi-Objective Simulation Optimization". In *Proceedings of the 2022 Winter Simulation Conference*, 3015–3026: IEEE Press.

Kazil, J., D. Masad, and A. Crooks. 2020. "Utilizing Python for Agent-Based Modeling: The Mesa Framework". In *Social, Cultural, and Behavioral Modeling (SBP-BRiMS)*, 308–317: Springer. Mesa is available at https://github.com/projectmesa/mesa.

Kendall, M. G. 1938. "A New Measure of Rank Correlation". *Biometrika* 30(1-2):81–93.

Lotka, A. J. 1925. "Elements of Physical Biology". *Science Progress in the Twentieth Century* 21(82):341–343.

Macal, C. M., and M. J. North. 2010. "Tutorial on Agent-Based Modelling and Simulation". *Journal of Simulation* 4(3):151–162.

OptTek Systems, Inc. 2007. "OptQuest Optimization Engine for Simulation". https://www.opttek.com/products/optquest/. Commercial simulation-optimization software; accessed March 2026.

Pangallo, M., J. D. Farmer, J. W. Sanders, and T. Galla. 2019. "Towards a Complete Classification of the Macroeconomic Consequences of Agent Heterogeneity". *Journal of Economic Dynamics and Control* 107:103716.

Pasupathy, R., and S. G. Henderson. 2011. "SimOpt: A Testbed for Simulation-Optimization Problems". In *Proceedings of the 2011 Winter Simulation Conference*, 4080–4090: IEEE Press.

Tesfatsion, L. 2006. "Agent-Based Computational Economics: A Constructive Approach to Economic Theory". In *Handbook of Computational Economics, Vol. 2*, edited by L. Tesfatsion and K. L. Judd, 831–880. North-Holland/Elsevier.

Volterra, V. 1926. "Variazioni e Fluttuazioni del Numero d'Individui in Specie Animali Conviventi". *Memorie della R. Accademia dei Lincei* 2:31–113.

Wilensky, Uri 1997. "NetLogo Wolf Sheep Predation Model". Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL. http://ccl.northwestern.edu/netlogo/models/WolfSheepPredation; Accessed March 2026.

Wilensky, Uri 1999. "NetLogo". Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL. http://ccl.northwestern.edu/netlogo/.

Zheng, S., A. Trott, S. Srinivasa, D. C. Parkes, and R. Socher. 2022. "The AI Economist: Taxation Policy Design via Two-level Deep Multiagent Reinforcement Learning". *Science Advances* 8(18):eabk2607.

Zitzler, E., and L. Thiele. 1998. "Multiobjective Optimization Using Evolutionary Algorithms—A Comparative Case Study". *Parallel Problem Solving from Nature*:292–301.

AUTHOR BIOGRAPHIES

RUIYU ZHANG is affiliated with the Department of Politics and Public Administration at The University of Hong Kong, Hong Kong SAR, China. Her e-mail address is Ruiyuzh@connect.hku.hk.

LIN NIE is affiliated with the Department of Applied Social Sciences at The Hong Kong Polytechnic University, Hong Kong SAR, China. His e-mail address is lin-apss.nie@polyu.edu.hk.

XIN ZHAO is affiliated with the Department of Applied Social Sciences at The Hong Kong Polytechnic University, Hong Kong SAR, China. His e-mail address is xinnn.zhao@connect.polyu.hk.